RANSMAP: Complete Information Guide

What it means

RanSMAP stands for **Ransomware Storage and Memory Access Patterns**. It is a modern, open-source dataset used in the cybersecurity field. It contains highly detailed logs of how both malicious ransomware and normal, benign computer applications interact with a computer's storage (like a hard drive) and memory (RAM).

How it occurs / How it is collected

The data inside RanSMAP is gathered using a specialized, low-level monitoring tool called a "thin hypervisor". Security researchers ran various famous ransomware samples (such as WannaCry, Ryuk, REvil, LockBit, Darkside, and Conti) alongside standard applications (like web browsers and Microsoft Office). While these programs were running, the thin hypervisor silently recorded exactly how they read and wrote data to the computer's memory and storage. This creates a highly accurate "behavioral fingerprint" of how ransomware operates when an infection occurs.

Main Purpose

The main purpose of RanSMAP is to train Artificial Intelligence (AI) and Machine Learning (ML) security systems. In the past, behavioral detectors using older datasets (like RanSAP, which only looked at storage) struggled to detect ransomware if it was running concurrently with normal programs. By providing a realistic, high-quality dataset of both storage and memory access patterns, RanSMAP allows developers to create much smarter, deep learning-based ransomware detectors. These detectors act as a strong behavioral defense layer that can stop a cyberattack even if traditional OS-level security software is compromised.

TYPES:

Ransomware generally falls into three main types:

Crypto-Ransomware: Encrypts valuable files but leaves the computer operating system running so the user can see the ransom note and pay.

Locker Ransomware: Completely locks the user out of the computer's interface (e.g., freezing the screen) rather than encrypting individual files.

Double Extortion: Before encrypting the files, the attackers copy (exfiltrate) the sensitive data to their own servers. If the victim refuses to pay for the decryption key, the attackers threaten to leak the stolen data to the public.

Identifying and Detecting Ransomware Files

Distinguishing a ransomware file from a normal file by sight alone is extremely difficult, as attackers intentionally disguise their malware. A ransomware file might look like a standard PDF, a Microsoft Word document containing malicious macros, or a legitimate software update.

To determine if a suspicious file is ransomware, you can use specialized detection sources and analysis methods:

File Extension and Naming: Ransomware often uses double extensions (e.g., invoice.pdf.exe) or disguises itself as system processes (e.g., svchost.exe running from the wrong folder).

Cloud Sandboxes (Safe Detonation): You can upload a suspicious file to online sandboxes like VirusTotal, Any.Run, or Hybrid Analysis. These platforms safely open the file in a secure, isolated cloud environment and analyze its behavior. If the file attempts to delete backups, modify the registry, or rapidly encrypt files, the platform flags it as ransomware.

Endpoint Detection and Response (EDR): Modern security software does not just look for known bad files; it monitors behavior. If a program suddenly starts reading and rewriting thousands of files per second (like the data tracked in the RanSMAP project), the EDR terminates the process.